

Kanishka Gunawardana

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Profile

Top-ranked Computer Engineering graduate (Rank 1/486, GPA 4.0/4.0) with strong interests in Computer Architecture, Neuromorphic Computing, and RISC-V SoC design, complemented by experience in embedded and intelligent systems. Dedicated to leveraging hardware-software co-design for energy-efficient edge computing, with demonstrated research excellence, leadership, and a collaborative mindset.

Education

University Of Peradeniya

Nov. 2021 – Aug. 2025

Undergraduate in B.Sc. Engineering(Hons.) Computer Engineering

GPA: 4.0/4.0, Rank: 1/486

Dharmaraja College, Kandy

Nov. 2006 – Aug. 2019

G.C.E. Advanced Level Examination

Z-score: 2.5661 (National Rank 149/19508)

Publications

SNAP-V: A RISC-V SoC with Configurable Neuromorphic Acceleration for Small-Scale Spiking Neural Networks

K. Gunawardana, S. Peeris, K. Rambukwella, T. Wanduragala, S. Jameel, R. Ragel, I. Nawinne
| *TNNLS (Under Review)*

- Proposed SNAP-V, a RISC-V-based neuromorphic SoC featuring two accelerator variants (bus-based Cerebra-S and NoC-based Cerebra-H) to provide a complete, resource-efficient platform tailored for small-scale edge SNNs.
- Demonstrated $9.46\times$ frequency improvement (96.24 MHz) in Cerebra-H, achieving 1.05 pJ/SOP synaptic energy in 45nm CMOS with 2.62% average hardware-software accuracy deviation across 80 configurations on MNIST.

Neuromorphic Architectures for Edge-Oriented Spiking Neural Networks: A Review

K. Gunawardana, S. Peeris, K. Rambukwella, R.G. Ragel, I. Nawinne | *JSA (Minor Revision)*

- Conducted a review of ASIC- and FPGA-based neuromorphic architectures, highlighting large-scale systems are underutilized for small-scale edge workloads while small-scale designs remain rare; analyzed software frameworks and event-driven communication strategies to advocate for SoC-centric platforms for energy-efficient deployment.

Optimized Multi-Processor System-on-Chip (MPSoC) Design for Low-Resource JPEG Encoding

K. Gunawardana, C. Adhikari, I. Nawinne | *ICAC 2024*

- Implemented an FPGA MPSoC JPEG encoder on Cyclone IV using Nios II/e cores; introduced lightweight custom hardware instructions and custom FIFO queues to offload compute-critical stages and reduce processor stalls.
- Achieved $2.8\times$ throughput improvement; superscalar options were evaluated but deprioritised.

A Configurable Neuromorphic Network-on-Chip Architecture for Spiking Neural Networks

I. Balasuriya, A. Karunanayake, T. Rathnayake, K. Gunawardana, S. Peeris, R. Ragel, M. Wickramasinghe, I. Nawinne | *MERCon 2026 (Submitted)*

- Executed rigorous experimental evaluations to benchmark the throughput, latency, and energy efficiency of the proposed NoC architecture; synthesized hardware performance data and co-authored the technical manuscript.

Research Experience

SNAP-V: A RISC-V SoC with Configurable Neuromorphic Acceleration for Small-Scale Spiking Neural Networks (Final Year Thesis) | [📄](#)

Nov. 2024 – Jul. 2025

- Designed a dual-core RISC-V SoC (MainCore + SpikeCore) with a hierarchical NoC neuromorphic accelerator integrated via the RoCC interface, utilizing Chisel and Verilog within the Chipyard framework.
- Built an end-to-end verification pipeline: RTL simulation (VCS), synthesis/timing (RTL Architect), power analysis (PrimePower), and FPGA prototyping (Xilinx Virtex-7); automated via Bash/Tcl/Python scripts.
- Validated hardware-software fidelity across 80 distinct network configurations on MNIST, incorporating comprehensive subsystem-level power attribution.
- Technology: Chisel, Chipyard, Verilog-HDL, Synopsys VCS/PrimePower/RTL Architect, Vivado

Technical Skills

Languages: Verilog HDL, Chisel/Scala, C/C++, ARM assembly, Python, Java, JavaScript, TypeScript, SQL, Ballerina.
Frameworks: Arduino, Express.js, Spring Boot, FastAPI, Node.js, React.js.
Libraries: OpenCV, NumPy, Pandas, Matplotlib, PyTorch, TensorFlow.
Hardware Design: Chipyard, RoCC, OpenRAM, Xilinx Vivado, Quartus II, Nios II, GTKWave.
Developer Tools: Linux, Git, Docker, AWS.
EDA & Verification: Synopsys Design Compiler, VCS, RTL Architect, PrimeTime, PrimePower, Tcl.

Experience

Teaching Assistant Aug. 2025 – Present
Department of Computer Engineering, University of Peradeniya

Instructing courses in computer architecture, digital design and embedded systems, including labs and evaluations.
Conducting neuromorphic accelerator research with the PeraMorphIQ Neuromorphic Research Group and mentoring research teams at the ESCAL Lab on on-chip learning, memory power optimizations, and system architecture.

Embedded Systems Engineering Intern (Part-Time) Nov. 2025 – Present
Attune Integration (Pvt) Ltd

Contributing to cutting-edge automation solutions by integrating AI-driven autonomous hardware with mobile software applications for a commercial, scalable robotic system, utilizing advanced hardware-software co-design principles.

Software Engineering Intern Jul. 2024 – Dec. 2024
WSO2 LLC, Colombo, Sri Lanka

Developed Ballerina integrations, including the OpenAI Finetunes Connector. Worked on ISO20022-to-SwiftMT message conversion using Ballerina for financial message interoperability, along with SaaS-based app design and development.

Undergraduate Teaching Assistant Jun. 2022 – Aug. 2025
Department of Computer Engineering, University of Peradeniya

Computing (GP106), Programming Methodology (CO222), Third Year Project (CO300) - Assisted in labs, quizzes, projects and course materials, supporting students in Python programming, C programming, DSA and project mentoring.

Awards and Achievements

J. B. Dissanayake Prize for Industrial Training | *Faculty of Engineering, University of Peradeniya* 2025
• Awarded for outstanding performance and excellence during the undergraduate industrial training program.

SLIoT Challenge 2023 | *Sri Lankan Biggest IOT Competition* | *Team: IMPAX* Mar. 2024
• 1st runners-up (Out of 100+ Teams) | *Organized by UOM in collaboration with SLT-MOBITEL and IESL*

MoraXtream 8.0 | *12 hour algorithmic programming competition* | *Team: Five4Five* Nov. 2023
• National Rank - 4 (Out of 400+ Teams) | *Organized by the IEEE Student Branch of the University of Moratuwa*

IEEEExtreme 17.0 | *24 hour algorithmic programming competition* | *Team: Five4Five* Nov. 2023
• Global Rank - 374 (Out of 16500+ participants), National Rank - 24 (Out of 330 Teams)

ACES Coders v10.0 | *12 hour algorithmic programming competition* | *Team: Five4Five* Oct. 2023
• National Rank - 12 (Out of 350+ participants) | *Organized by the ACES*

Selected Projects

RV32IM 5-stage Pipeline Processor | *Group* |  Dec. 2024 – Jul. 2025
• Designed a 5-stage pipelined RISC-V RV32IM processor using Verilog HDL and prototyped it on a Xilinx Virtex-7 FPGA, featuring in-order hazard handling and AXI-based memory integration for SoC compatibility.
• Established an automated CI/CD verification pipeline via GitHub Actions to execute static timing and power analysis using Synopsys RTLA and PrimePower, achieving a maximum frequency of 142 MHz at 0.197 mW.
• Technology: **Verilog HDL, Synopsys DC, VCS, RTLA, PrimePower, GTKWave, GitHub**

Impact Tracking System for Athletes (3YP) | *Group* |   Nov. 2023 – Mar. 2024

- Developed a real-time head impact monitoring system for contact sports using wearable devices and desktop dashboards, facilitating prompt concussion identification, post-session data transmission, and comprehensive analytics for player safety and informed decision-making.
- Contributions: Led hardware and firmware design and development of wearable devices, developed the centralized hub and WIFI local communication, contributed to backend API, and deployed the system on AWS EC2.
- Technologies: C++, ESP32, Raspberry Pi, MQTT, Python, Express.js, MongoDB, AWS

Field-Based Approach for Quantifying Plant Leaf Color | *Group* |   Aug. 2023 – Nov. 2023

- Developed a mobile application with a backend that utilizes Image Processing and Computer Vision to objectively quantify plant leaf colour by analyzing information extracted from captured leaf images.
- Contributions: Developed the backend API for image analysis using FastAPI and contributed to image preprocessing, including image segmentation with a Mask R-CNN model fine-tuned for leaf segmentation.
- Technology: Python, OpenCV, Pytorch, FastAPI

Obstacle Robot Swarm for Swarm Robotic Project | *Group* |   Feb. 2023 - Nov. 2024

- Led the development and firmware updates of obstacle-avoiding robots equipped with navigation and collision avoidance algorithms, utilizing a gyroscope and accelerometer for the swarm robotics platform.
- Integrating obstacle robots with the existing swarm platform, enabling studies of dynamic obstacle scenarios.
- Technology: Arduino, Python, Java, MQTT, OpenCV

8-bit Single-cycle Processor |  Mar. 2023 – Jun. 2023

- Designed and implemented an 8-bit single-cycle processor architecture in Verilog HDL with instruction and data caches, featuring a memory unit and a MIPS-inspired ISA supporting arithmetic, logic, and control operations.
- Developed a comprehensive testbench for verification and waveform analysis using GTKWave tool.
- Technology: Verilog-HDL, GTKWave

Certificates

Machine Learning Specialization - Stanford University & DeepLearning.AI (Coursera) Sep. 2023

- Supervised Machine Learning: Regression and Classification
- Unsupervised Learning, Recommenders, Reinforcement Learning
- Advanced Learning Algorithms

Extra-Curricular Activities

Head of Web Development - Robotics Society, University of Peradeniya Sep. 2023 – Aug. 2024

Volunteering Project Nenathambara (Rural education outreach initiative) Sep. 2023 – Jul. 2024

Executive Committee Member - Robotics Society, University of Peradeniya Dec. 2022 – Sep. 2023

Member of Rotaract Club of University of Peradeniya Dec. 2021 – Dec. 2023

References

Dr. Isuru Nawinne | *isurunawinne@eng.pdn.ac.lk*

Senior Lecturer, Department of Computer Engineering, Faculty of Engineering, University of Peradeniya, Sri Lanka.

Prof. Roshan G. Ragel | *roshanr@eng.pdn.ac.lk*

Professor, Department of Computer Engineering, Faculty of Engineering, University of Peradeniya, Sri Lanka.